```python
import pandas as pd

# Load the dataset
df = pd.read_csv('c107fa55-45be-4f9c-8f61-088e88d1fb0c.csv')

# Convert TV from text to numbers
tv_map = {'Low': 1, 'Medium': 2, 'High': 3}
df['TV'] = df['TV'].map(tv_map)

# Keep only numeric columns
numeric_cols = ['TV', 'Radio', 'Social Media', 'Sales']
df = df[numeric_cols]

# Check if it worked
print(df.head())
print("\nData types:")
print(df.dtypes)
```

```
   TV      Radio  Social Media       Sales
0   1   3.518070      2.293790   55.261284
1   1   7.756876      2.572287   67.574904
2   3  20.348988      1.227180  272.250108
3   2  20.108487      2.728374  195.102176
4   3  31.653200      7.776978  273.960377

Data types:
TV                int64
Radio           float64
Social Media    float64
Sales           float64
dtype: object
```

```python
# ============================================
# MULTIPLE LINEAR REGRESSION
# MULTI-CHANNEL MARKETING ANALYSIS
# COMPLETE SUBMISSION
# ============================================

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
from statsmodels.stats.outliers_influence import variance_inflation_factor
from scipy import stats

plt.style.use('seaborn-v0_8-whitegrid')
sns.set_palette("husl")

# ============================================
# TASK 1: LOAD DATASET & CONVERT TEXT
# ============================================

print("="*60)
```

```
print("TASK 1: LOAD DATASET & EXPLORATORY DATA ANALYSIS")
print("="*60)

# Load dataset
df = pd.read_csv('c107fa55-45be-4f9c-8f61-088e88d1fb0c.csv')

# Convert TV from text to numbers
tv_map = {'Low': 1, 'Medium': 2, 'High': 3}
df['TV'] = df['TV'].map(tv_map)

# Keep only numeric columns
numeric_cols = ['TV', 'Radio', 'Social Media', 'Sales']
df = df[numeric_cols]

print(f"\nDataset shape: {df.shape}")
print("\nFirst 5 rows:")
print(df.head())

print("\nBasic statistics:")
print(df.describe())

# ============================================
# TASK 2: VISUALIZATIONS
# ============================================

print("\n" + "="*60)
print("EXPLORATORY DATA ANALYSIS - VISUALIZATIONS")
print("="*60)

# Histograms
fig, axes = plt.subplots(2, 2, figsize=(12, 10))
axes = axes.flatten()

for i, col in enumerate(['TV', 'Radio', 'Social Media', 'Sales']):
    axes[i].hist(df[col], bins=30, edgecolor='black', alpha=0.7)
    axes[i].set_title(f'Distribution of {col}')
    axes[i].set_xlabel(col)
    axes[i].set_ylabel('Frequency')

plt.tight_layout()
plt.show()

# Pairplot
sns.pairplot(df)
plt.show()

# Scatter plots
fig, axes = plt.subplots(1, 3, figsize=(15, 5))

axes[0].scatter(df['TV'], df['Sales'], alpha=0.6)
axes[0].set_xlabel('TV Spend')
axes[0].set_ylabel('Sales')
axes[0].set_title('TV vs Sales')

axes[1].scatter(df['Radio'], df['Sales'], alpha=0.6, color='green')
axes[1].set_xlabel('Radio Spend')
axes[1].set_ylabel('Sales')
axes[1].set_title('Radio vs Sales')

axes[2].scatter(df['Social Media'], df['Sales'], alpha=0.6, color='orange')
```

```
axes[2].set_xlabel('Social Media Spend')
axes[2].set_ylabel('Sales')
axes[2].set_title('Social Media vs Sales')

plt.tight_layout()
plt.show()

# Correlation heatmap
plt.figure(figsize=(8, 6))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm', center=0)
plt.title('Correlation Matrix')
plt.show()

# ===========================================
# TASK 3: CHECK MULTICOLLINEARITY (VIF)
# ===========================================

print("\n" + "="*60)
print("TASK 3: CHECK MULTICOLLINEARITY (VIF)")
print("="*60)

correlations = df.corr()['Sales'].sort_values(ascending=False)
print("\nCorrelation with Sales:")
print(correlations)

X = df[['TV', 'Radio', 'Social Media']]
vif_data = pd.DataFrame()
vif_data["Variable"] = X.columns
vif_data["VIF"] = [variance_inflation_factor(X.values, i) for i in range(X.sha

print("\nVariance Inflation Factor (VIF):")
print(vif_data)

# ===========================================
# TASK 4: BUILD MULTIPLE LINEAR REGRESSION MODEL
# ===========================================

print("\n" + "="*60)
print("TASK 4: MULTIPLE LINEAR REGRESSION MODEL")
print("="*60)

X = df[['TV', 'Radio', 'Social Media']]
y = df['Sales']
X_const = sm.add_constant(X)
model = sm.OLS(y, X_const).fit()
print(model.summary())

# ===========================================
# TASK 5: DIAGNOSTIC PLOTS
# ===========================================

print("\n" + "="*60)
print("TASK 5: DIAGNOSTIC PLOTS (ASSUMPTION CHECKS)")
print("="*60)

fitted = model.fittedvalues
residuals = model.resid

fig, axes = plt.subplots(2, 2, figsize=(12, 10))
```

```
axes[0, 0].scatter(fitted, residuals, alpha=0.6)
axes[0, 0].axhline(y=0, color='red', linestyle='--')
axes[0, 0].set_xlabel('Fitted Values')
axes[0, 0].set_ylabel('Residuals')
axes[0, 0].set_title('Residuals vs Fitted (Linearity)')

stats.probplot(residuals, dist="norm", plot=axes[0, 1])
axes[0, 1].set_title('Q-Q Plot (Normality)')

std_res = np.sqrt(np.abs(residuals))
axes[1, 0].scatter(fitted, std_res, alpha=0.6)
axes[1, 0].set_xlabel('Fitted Values')
axes[1, 0].set_ylabel('√|Residuals|')
axes[1, 0].set_title('Scale-Location (Homoscedasticity)')

axes[1, 1].hist(residuals, bins=30, edgecolor='black', alpha=0.7)
axes[1, 1].set_xlabel('Residuals')
axes[1, 1].set_ylabel('Frequency')
axes[1, 1].set_title('Histogram of Residuals')

plt.tight_layout()
plt.show()

# ============================================
# TASK 6: INTERPRETATION
# ============================================

print("\n" + "="*60)
print("TASK 6: INTERPRETATION OF COEFFICIENTS")
print("="*60)

print(f"\n📊 R-squared: {model.rsquared:.4f}")
print(f"📊 Adjusted R-squared: {model.rsquared_adj:.4f}")

print("\n📈 Linear Equation:")
print(f"Sales = {model.params[0]:.4f} + {model.params[1]:.4f}*TV + {model.parar

print("\n📊 Interpretation (holding all other variables constant):")
print(f"   • For each ₦1 increase in TV spend, Sales increases by ₦{model.para
print(f"   • For each ₦1 increase in Radio spend, Sales increases by ₦{model.p
print(f"   • For each ₦1 increase in Social Media spend, Sales increases by ₦{

# ============================================
# TASK 7: BUSINESS RECOMMENDATION
# ============================================

print("\n" + "="*60)
print("TASK 7: BUSINESS RECOMMENDATION")
print("="*60)

coefs = model.params[1:]
max_coef = coefs.idxmax()
max_val = coefs.max()

print(f"\n💡 The strongest predictor of Sales is: {max_coef}")
print(f"   Each ₦1 spent on {max_coef} generates ₦{max_val:.2f} in sales.\n")

print("✅ RECOMMENDATION:")
print(f"   Allocate the majority of the marketing budget to {max_coef.upper()}
```

```
print("\n" + "="*60)
print("PROJECT COMPLETE - ALL REQUIREMENTS MET")
print("="*60)
```

```
============================================================
TASK 1: LOAD DATASET & EXPLORATORY DATA ANALYSIS
============================================================

Dataset shape: (572, 4)

First 5 rows:
   TV      Radio  Social Media       Sales
0   1   3.518070      2.293790   55.261284
1   1   7.756876      2.572287   67.574904
2   3  20.348988      1.227180  272.250108
3   2  20.108487      2.728374  195.102176
4   3  31.653200      7.776978  273.960377

Basic statistics:
               TV       Radio  Social Media       Sales
count  572.000000  572.000000    572.000000  572.000000
mean     1.938811   17.520616      3.333803  189.296908
std      0.799363    9.290933      2.238378   89.871581
min      1.000000    0.109106      0.000031   33.509810
25%      1.000000   10.699556      1.585549  118.718722
50%      2.000000   17.149517      3.150111  184.005362
75%      3.000000   24.606396      4.730408  264.500118
max      3.000000   42.271579     11.403625  357.788195

============================================================
EXPLORATORY DATA ANALYSIS - VISUALIZATIONS
============================================================
```